



Fullstack Web Development

Session 1 | Week 1/Day 2 | 120 min

HTML5 - Part02

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HTML Forms

What is an HTML Form?

HTML Forms

What is an HTML Form?

- An **HTML form** allows users to **enter data** on a webpage, which can be sent to a server for processing.

- It is created using the <form> element:

```
<form>
    .
     form elements
    .
</form>
```

HTML Forms

How the <form> **Element Works**

- Container for input elements such as:
  - Text fields
  - Password fields
  - Checkboxes
  - Radio buttons
  - Buttons (Submit / Reset)

- Wraps all the input elements together

```
<form action="submit.php" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="user_name">

    <label for="email">Email:</label>
    <input type="email" id="email" name="user_email">

    <input type="submit" value="Submit">
</form>
```

HTML Forms

Important Attributes of <form>

Attribute	Description
action	URL where the form data is sent for processing
method	How the data is sent: GET or POST

```
<form action="submit.php" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="user_name">

    <label for="email">Email:</label>
    <input type="email" id="email" name="user_email">

    <input type="submit" value="Submit">
</form>
```

HTML Forms

The action **Attribute in HTML Forms**

What It Does?
The **action attribute** specifies **where the form data should be sent** when the user clicks the **Submit** button.

- **Example:** Send the data to a server file called **submit.php**
- If action is **omitted**, the data is sent to the **current page** by default

HTML Forms

The action Attribute in HTML Forms

- **Example Form:**

```
<form action="/action_page.php" method="POST">
    <label for="fname">First Name:</label><br>
    <input type="text" id="fname" name="fname" value="John"><br>

    <label for="lname">Last Name:</label><br>
    <input type="text" id="lname" name="lname" value="Doe"><br><br>

    <input type="submit" value="Submit">
</form>
```

When the user clicks Submit, the data (fname=John&lname=Doe) is sent to /action_page.php.

The server file can then process this data (save to database, send email, ...)

HTML Forms

The method **Attribute in HTML Forms**

What It Does?
The **method attribute** defines **how the form data is sent to the server.**

- **Two Common Methods**

Method	How It Works	Use Case
GET	Data is sent as part of the URL (query string: name=value&name2=value2)	Searching, simple data retrieval, bookmarking URLs
POST	Data is sent in the **HTTP request body** (not visible in URL)	Submitting sensitive information, large forms, login credentials

HTML Forms

The method **Attribute in HTML Forms**

- **Example** GET Method:

```
<form action="/submit.php" method="GET">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name">
    <input type="submit" value="Submit">
</form>
```

If user enters “John”, the URL becomes: /submit.php?name=John

Data is visible in the URL

**Default method is GET if you don’t specify it*

HTML Forms

The method **Attribute in HTML Forms**

- **Example** POST Method:

```
<form action="/submit.php" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name">
    <input type="submit" value="Submit">
</form>
```

Data is sent in the HTTP request body

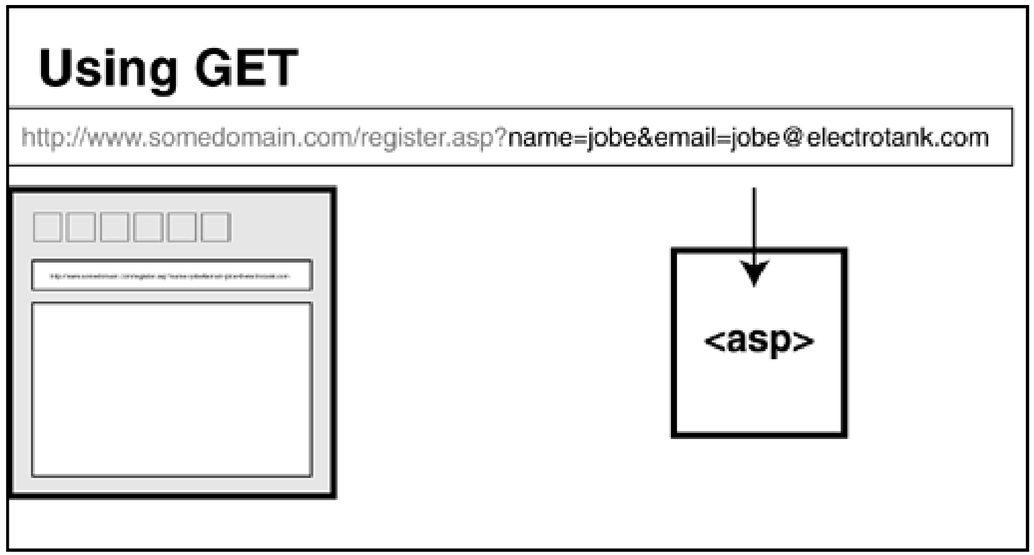
Not visible in the URL

More secure for sensitive information

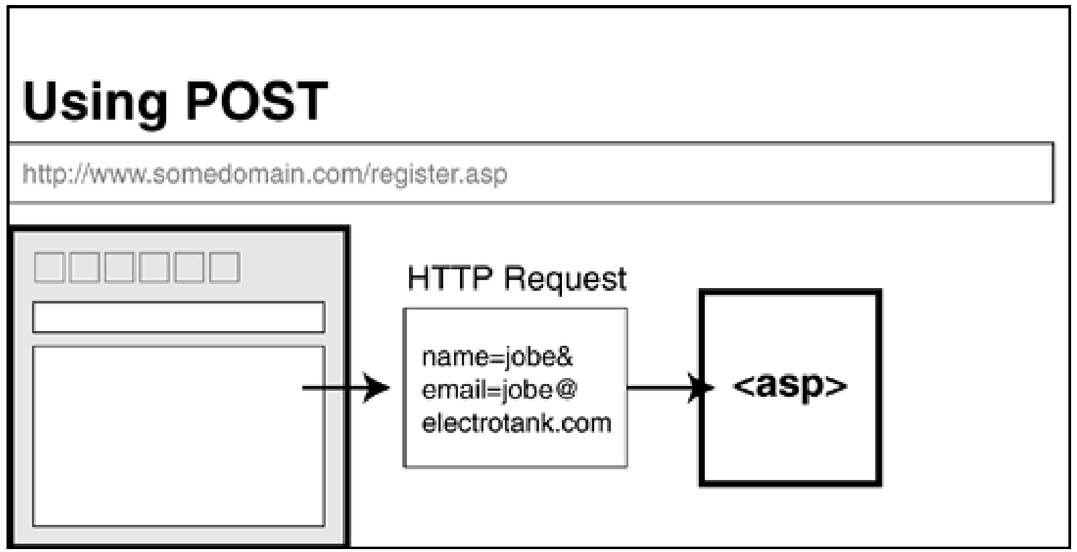
HTML Forms

The method **Attribute in HTML Forms**

- **Example** :



Data is visible in the URL



Not visible in the URL

HTML Forms

HTML Form Attributes : Submission Behavior

Attribute	Description	Example
action	URL where form data is sent	<form action="/submit.php">
method	HTTP method used (GET or POST)	<form method="POST">
enctype	How the data is encoded (only for POST)	<form enctype="multipart/form-data">
accept-charset	Character encoding for form submission	<form accept-charset="UTF-8">

HTML Forms

HTML Form Attributes : User Experience

Attribute	Description	Example
autocomplete	Enables/disables automatic field filling	<form autocomplete="off">
novalidate	Skips HTML validation on submission	<form novalidate>

HTML Forms

HTML Form Attributes : Form Identification & Linking

Attribute	Description	Example
name	Name of the form (used in scripts)	<form name="myForm">
rel	Relationship between linked resource & current page	<form rel="noopener">
target	Where to display the response	<form target="_blank">

HTML Forms

HTML <form> **Elements**

The **<form>** element can contain **various input and control elements.** Here’s a breakdown:

Element	Description	Example
<input>	Single-line input field (text, password, checkbox, ...)	<input type="text" name="username">
<label>	Defines a label for an input (improves accessibility)	<label for="username">Username: </label>
<select>	Drop-down list	<select><option>Option 1</option> </select>

HTML Forms

HTML <form> **Elements**

The **<form>** element can contain **various input and control elements.** Here's a breakdown:

Element	Description	Example
<textarea>	Multi-line text input	<textarea name="message"> </textarea>
<button>	Clickable button (submit, reset, or custom)	<button type="submit">Send</button>
<fieldset>	Groups related form elements	<fieldset><legend>Personal Info</legend>...</fieldset>
<legend>	Caption for a <fieldset>	<legend>Personal Info</legend>

HTML Forms

HTML <form> **Elements**

The **<form>** element can contain **various input and control elements.** Here's a breakdown:

Element	Description	Example
<datalist>	Predefined list of options for an <input>	<input list="colors"><datalist id="colors"><option value="Red"> </datalist>
<output>	Displays calculation results	<output name="result"></output>
<option>	Defines an option in <select> or <datalist>	<option value="1">One</option>
<optgroup>	Groups related options in a drop-down list	<optgroup label="Fruits"> <option>Apple</option></optgroup>

HTML Input Types

What is the main purpose of the **<input>** element in a form?

- The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
text	One-line text input	<input type="text" name="username">
password	One-line input for sensitive text; characters are hidden	<input type="password" name="pwd">
submit	Sends form data to the server	<input type="submit" value="Send">
reset	Resets all form fields to default values	<input type="reset" value="Clear">
radio	Select one option from multiple choices	<input type="radio" name="gender" value="male">

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
checkbox	Select multiple options	<input type="checkbox" name="hobby" value="reading">
button	Simple push button (can be programmed with JavaScript)	<input type="button" value="Click me">
file	Upload a file from user’s device	<input type="file" name="resume">
image	Graphical submit button (clickable image)	<input type="image" src="submit.png" alt="Submit">
color	Input field for selecting a color	<input type="color" name="favcolor">

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
date	Input field to select a date	<input type="date" name="birthday">
datetime-local	Input field for date and time (no time zone)	<input type="datetime-local" name="meeting">
email	Input field for email addresses	<input type="email" name="user_email">
month	Input field to select month and year	<input type="month" name="expiry">
number	Input field to enter a number	<input type="number" name="quantity" min="1" max="10">

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
url	Input field to enter a website URL	<input type="url" name="website">
week	Input field for week and year	<input type="week" name="weeknum">
search	Single-line field for search queries	<input type="search" name="searchbox">
tel	Input field for telephone numbers	<input type="tel" name="phone">

HTML Input Types

“I want to create a form where users can select one favorite color or multiple hobbies. Which input type should I use?”

text

checkbox

radio

submit

HTML Input Types Validation

HTML provides several **attributes to validate or control input fields:**

Attribute	Description	Example
maxlength	Maximum number of characters allowed	<input type="text" name="username" maxlength="10">
min	Minimum value allowed (for number/date inputs)	<input type="number" name="age" min="18">
max	Maximum value allowed (for number/date inputs)	<input type="number" name="age" max="99">
readonly	Field cannot be edited by the user	<input type="text" name="id" value="123" readonly>

HTML Input Types Validation

HTML provides several **attributes to validate or control input fields:**

Attribute	Description	Example
required	Field must be filled before form submission	<input type="email" name="email" required>
disabled	Field is unusable and unclickable	<input type="text" name="promo" disabled>

HTML Layout (Semantic Elements)

HTML5 introduced **semantic layout elements** to describe the structure of a webpage clearly.

These elements help:

- Organize content
- Improve readability
- Improve SEO
- Improve accessibility

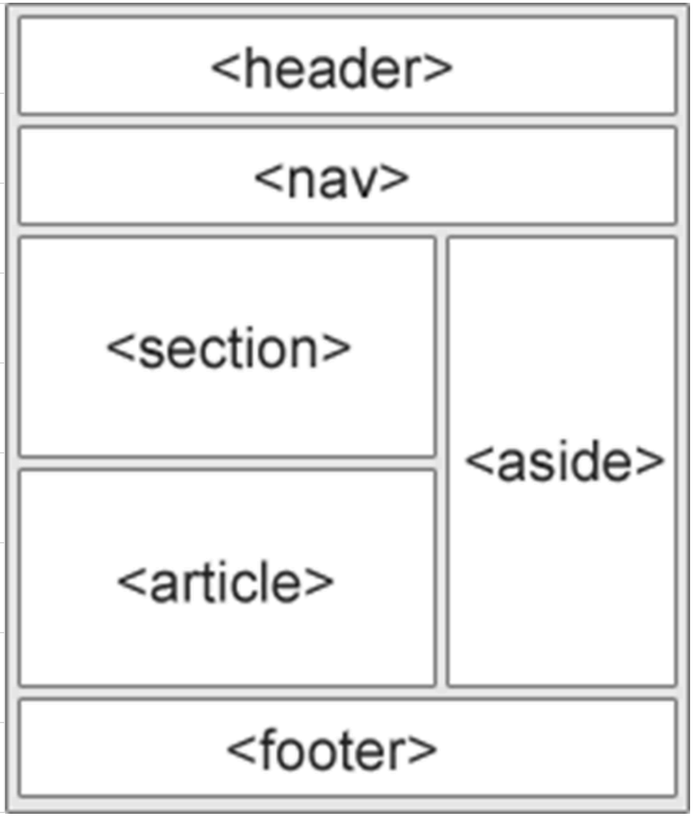
HTML Layout (Semantic Elements)

Common **Layout Elements**

HTML Layout (Semantic Elements)

Common **Layout Elements**

Element	Description
<header>	Field must be filled before form submission
<nav>	Field is unusable and unclickable
<section>	Groups related content into sections
<article>	Self-contained content (blog post, news article)
<aside>	Sidebar content (ads, related links, info box)
<footer>	Bottom section (copyright, contact info)
<details>	Expandable content block
<summary>	Visible heading for <details>



HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Frameworks
Pre-built CSS libraries that help build layouts quickly.

- Examples:
  - Bootstrap
  - Tailwind CSS
  - Foundation
- They provide:
  - Grid systems
  - Responsive design
  - Ready-made components









HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Float (Older Technique)

- Originally used to wrap text around images
- Later used for page layouts
- Difficult to manage in complex layouts

Considered outdated for modern layouts.

```
.sidebar {
    float: right;
}
```

HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Flexbox (Modern Layout)

- One-dimensional layout system
- Arranges items in a row or column
- Easy alignment and spacing
- Great for navbars, cards, small components

Best for Component-level layout.

```
.container {
    display: flex;
}
```

HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Grid (Most Powerful)

- Two-dimensional layout system
- Controls rows and columns
- Perfect for full page layout

Best for Complex page structure.

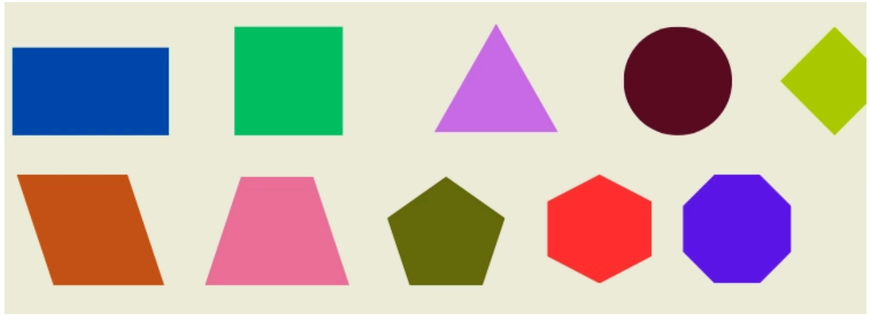
```
.container {
    display: grid;
    grid-template-columns: 1fr 1fr;
}
```

HTML5 Graphics

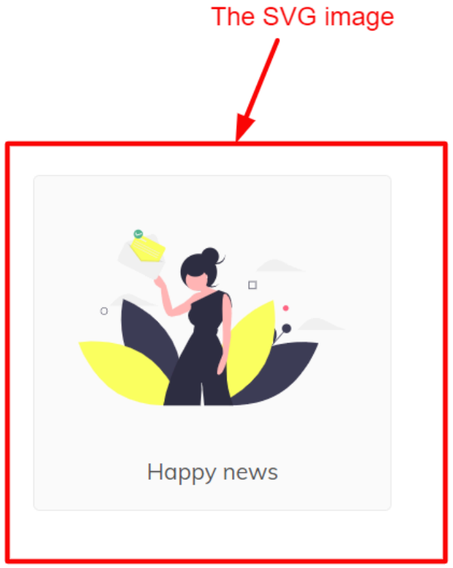
HTML5 provides **two main graphic** technologies:

- <canvas>
- <svg>

But they are used in different situations.



<canvas>



HTML5 Graphics – <canvas>

What is <canvas>**?**

The **<canvas>** element is used to **draw graphics dynamically** using JavaScript.

It can be used to create:

- Charts
- Games
- Animations
- Drawing applications
- Image manipulation

```
<canvas id="myCanvas" width="200" height="100"
    style="border:1px solid #000000;">
</canvas>
```

This only creates a blank rectangle.



Type : Pixel

<canvas> is just a blank drawing area.
Without JavaScript, it will only display an empty box.

HTML5 Graphics – SVG

What is SVG**?**

SVG (Scalable Vector Graphics) defines graphics using **XML-based markup.**

It is used to create:

- Logos
- Icons
- Charts
- Illustrations
- Scalable graphics

```
<svg width="100" height="100">
    <circle cx="50" cy="50" r="40" stroke="green" stroke-width="4" fill="yellow" />
    Sorry, your browser does not support inline SVG.
</svg>
```

Type : Vector

- *<svg> → container*
- *<circle> → draws a circle*
- *cx → center X position*
- *cy → center Y position*
- *r → radius*
- *stroke → border color*
- *fill → inside color*





THANK YOU

For your attention !

